

DFT calculation

June 13, 2026

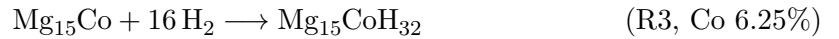
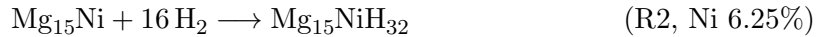
1 Destabilisation of MgH_2 and Mg_2NiH_4 by substitutional Ni and Co dopants: a first-principles study

Magnesium-based hydrides are leading materials for solid-state hydrogen storage, with high gravimetric capacity and low cost. Pure MgH_2 holds 7.6 % hydrogen by mass, but its large formation enthalpy ($\Delta H_f = -74.5 \text{ kJ mol}^{-1} \text{ H}_2$ [1, 2]) makes the hydride too stable for practical use, requiring dehydrogenation near 300 °C. The intermetallic Mg_2Ni forms a less stable hydride, Mg_2NiH_4 ($\Delta H = -64.4 \text{ kJ mol}^{-1} \text{ H}_2$ [3]), which desorbs more easily but stores less (~ 3.6 %). An effective additive destabilises the hydride, raising ΔH toward zero and lowering the desorption temperature.

We use density-functional theory (DFT) to compute the hydride formation enthalpy for a single substitutional Ni or Co atom in each host, quantifying the destabilisation produced by each dopant and, for Co in Mg_2Ni , its dependence on the substitution site.

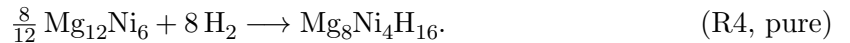
1.1 Methods

We evaluate six hydrogenation reactions, three per host, on a common per- H_2 basis. For the Mg host, reactant and product supercells have identical metal content, so ΔH follows from the balanced reaction:



The supercells are $2 \times 2 \times 2$ replications of rutile MgH_2 (product, 48 atoms) and hexagonal close-packed Mg (reactant, 16 metal sites), with one site replaced by the dopant: 6.25 at %, equivalent to 13.87 wt % (Ni) or 13.92 wt % (Co) of the metal. All Mg sites are symmetry-equivalent, so the placement is unique.

For the Mg_2Ni host the pristine reaction uses the 18-atom hexagonal Mg_2Ni cell and the 28-atom monoclinic ($C2/c$) Mg_2NiH_4 cell:



The doped 18- and 28-atom cells differ in metal content, so doped enthalpies use the dilute-defect limit,

$$\Delta\Delta H = \frac{1}{n_{\text{H}_2}} [(E_{\text{hyd}}^{\text{doped}} - E_{\text{hyd}}^{\text{pure}}) - (E_{\text{met}}^{\text{doped}} - E_{\text{met}}^{\text{pure}})] \times 1312.75 \text{ kJ mol}^{-1} \text{ Ry}^{-1}, \quad (1)$$

in which the Mg and dopant chemical potentials cancel. We treat Ni at an Mg site (R5), Co at an Mg site (R6), and Co at a Ni site (R6'), the last because Co favours the Ni site in Mg_2Ni ($E_{\text{Mg} \rightarrow \text{Ni}} = -0.53 \text{ eV}$ [4]). Figures 1 and 2 show the relaxed product structures.

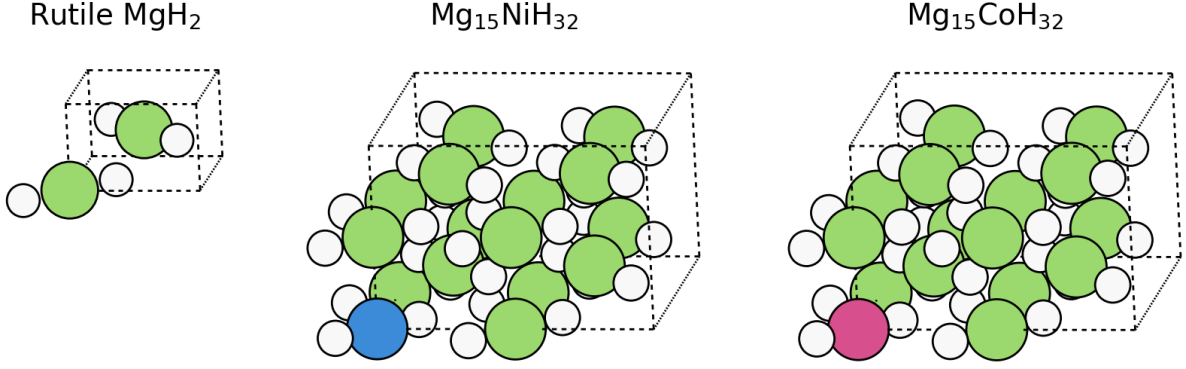


Figure 1: Relaxed product structures for the Mg host: pristine rutile MgH_2 (left) and the Ni- and Co-substituted hydrides $\text{Mg}_{15}\text{NiH}_{32}$ and $\text{Mg}_{15}\text{CoH}_{32}$ (right) at 6.25 at %. Atom colours: Mg (green), H (white), Ni (blue), Co (pink).

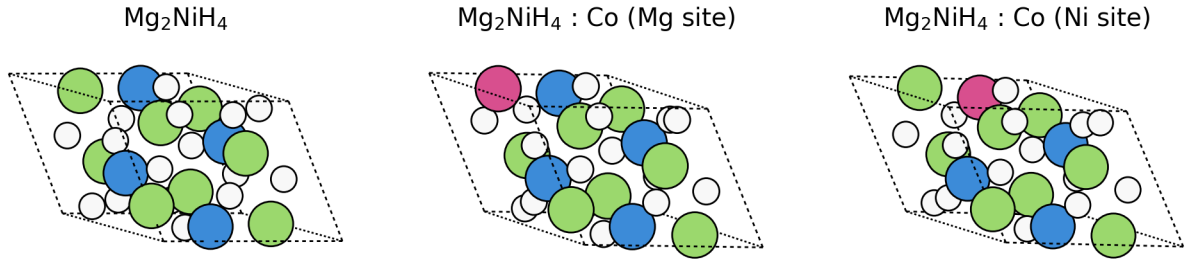


Figure 2: Relaxed product structures for the Mg_2Ni host: pristine Mg_2NiH_4 (left), and Co substituted at an Mg site (centre, R6) and at a Ni site (right, R6'). Co favours the Ni site, where the destabilising effect is negligible.

All calculations use Quantum ESPRESSO [5] with the Perdew–Burke–Ernzerhof (PBE) functional and Grimme-D3 dispersion [6]. Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (charge density); pseudopotentials are projector augmented-wave for Mg and H, ultrasoft for Ni and Co. Brillouin-zone integration uses Γ -centred grids at $\sim 0.03 \text{ \AA}^{-1}$ ($12 \times 12 \times 8$ for the Mg primitive cell, $8 \times 8 \times 12$ for rutile MgH_2 , $6 \times 6 \times 4$ for the Mg supercell, $4 \times 4 \times 6$ for the hydride supercell, $8 \times 8 \times 4$ for Mg_2Ni , and $8 \times 8 \times 6$ for Mg_2NiH_4). Metals use cold smearing (0.01 Ry); the isolated H_2 molecule uses fixed occupations. The self-consistent field is converged to 1×10^{-9} Ry, and geometries relaxed until forces fall below $1 \times 10^{-4} \text{ Ry } a_0^{-1}$. No Hubbard U is applied, so all six reactions share one energy scale. The formation enthalpy per H_2 is

$$\Delta H = \frac{1}{n_{\text{H}_2}} [E(\text{hydride}) - E(\text{metal}) - n_{\text{H}_2} E(\text{H}_2)] \times 1312.75 \text{ kJ mol}^{-1} \text{ Ry}^{-1}. \quad (2)$$

Each cell is evaluated in its electronic ground state, confirmed by spin-polarised single-point calculations. Ni is nonmagnetic in every environment; spin-polarised relaxation of the Ni-substituted metal gives zero moment. Co carries an environment-dependent moment: $1.84 \mu_{\text{B}}$ in Mg_{15}Co , $2.83 \mu_{\text{B}}$ and $1.45 \mu_{\text{B}}$ at the Mg and Ni sites of metallic Mg_2Ni , and $1.18 \mu_{\text{B}}$ in $\text{Mg}_{15}\text{CoH}_{32}$. It is quenched ($\leq 0.4 \mu_{\text{B}}$) only at the hydrogen-coordinated sites of Mg_2NiH_4 . Cells with a finite Co moment are relaxed spin-polarised. This shifts $\Delta\Delta H$ from $+4.70$ to 4.09 kJ mol^{-1} (Mg:Co) and from $+13.28$ to $15.03 \text{ kJ mol}^{-1}$ ($\text{Mg}_2\text{Ni:Co}$, Mg site), without changing either sign.

1.2 DFT results

Table 1 lists the formation enthalpies. Each dopant destabilises both hosts at the Mg site: the four Mg-site $\Delta\Delta H$ values are positive and exceed the numerical uncertainty by an order of

magnitude. Ni is the stronger destabiliser ($\Delta\Delta H = +11.86$ and $14.43 \text{ kJ mol}^{-1}$), and Mg:Ni retains the full 7.1 wt % capacity of magnesium.

	Pathway	ΔH / $\text{kJ mol}^{-1} \text{ H}_2$	$\Delta\Delta H$	Effect
R1	pure Mg	−63.75	—	baseline
R2	Mg : Ni (6.25%)	−51.89	11.86	destabilises
R3	Mg : Co (6.25%)	−59.66	4.09	destabilises
R4	pure Mg_2Ni	−66.97	—	baseline
R5	Mg_2Ni : Ni	−52.54	14.43	destabilises
R6	Mg_2Ni : Co (Mg site)	−51.94	15.03	destabilises
R6′	Mg_2Ni : Co (Ni site)	−68.71	−1.74	$\approx$ neutral

Table 1: Hydrogenation enthalpy per absorbed H_2 for the six pathways. A less-negative ΔH (positive $\Delta\Delta H$) indicates a less stable hydride, which is easier to dehydrogenate. $\Delta\Delta H$ is referenced to the corresponding pure host (R1 for the Mg host, R4 for the Mg_2Ni host).

The exception is Co at its preferred Ni site (R6′), which is marginally stabilising ($\Delta\Delta H = -1.74 \text{ kJ mol}^{-1}$). This is an order of magnitude smaller than the Mg-site shifts, comparable to the $\sim 1.4 \text{ kJ mol}^{-1}$ variation across the three inequivalent Mg sites of Mg_2NiH_4 [4], and well below the $\sim 10 \text{ kJ mol}^{-1}$ zero-point correction, so Co at the Ni site is effectively inert. The benefit of Co in Mg_2Ni thus depends on site occupancy: the Mg site destabilises strongly, as in non-equilibrium ball-milled material, while the equilibrium Ni site does not.

The pure-Mg baseline ($\Delta H = -63.75 \text{ kJ mol}^{-1} \text{ H}_2$) lies $\sim 11 \text{ kJ mol}^{-1}$ above experiment, reflecting the known GGA under-binding of ionic MgH_2 (Er et al. obtain $-63.7 \text{ kJ mol}^{-1}$ with PW91 [7]; Pozzo and Alf  require quantum Monte Carlo to match experiment [8]). The pure- Mg_2Ni baseline ($-66.97 \text{ kJ mol}^{-1}$) agrees with experiment [3] to within 2.6 kJ mol^{-1} ; the hydrogen in Mg_2NiH_4 forms covalent $[\text{NiH}_4]^{4-}$ complexes [4] that the GGA describes more accurately than the ionic H^- of MgH_2 . This host-dependent error ($\sim 13 \text{ kJ mol}^{-1}$) prevents the absolute enthalpies from reproducing the experimental host ordering, so we draw no cross-host conclusions from absolute values. The within-host differences $\Delta\Delta H$ cancel this error and are robust; the same cancellation underlies the GGA survey of Yoon et al. [4], whose relative predictions reproduce plateau-pressure trends for seven solutes in MgH_2 and ten in Mg_2Ni . For an absolute estimate, the shift can be referenced to the experimental host enthalpy, $\Delta H \approx \Delta H_{\text{exp}} + \Delta\Delta H_{\text{DFT}}$, giving $-62.6 \text{ kJ mol}^{-1}$ for Mg:Ni and $-50.0 \text{ kJ mol}^{-1}$ for Mg_2Ni :Ni, with estimated 1-bar desorption temperatures of 207°C and 110°C (van ’t Hoff, $\Delta S = 130.5 \text{ J mol}^{-1} \text{ K}^{-1}$).

A single Ni or Co atom at an Mg site destabilises both MgH_2 and Mg_2NiH_4 , lowering the thermodynamic barrier to hydrogen release. Ni is the stronger, more reliable catalyst for both hosts, and Mg:Ni preserves the high capacity of magnesium. Co is effective in Mg and at the Mg site of Mg_2Ni , but inert at its preferred Ni site. These results give a thermodynamic basis for the improved dehydrogenation of Ni- and Co-doped magnesium hydrides. Two limitations apply: the calculations address thermodynamics only, not the activation barriers behind the additives’ kinetic role; and zero-point energy, omitted here, amounts to $\sim 15 \text{ kJ mol}^{-1} \text{ H}_2$ for these hydrides [7] but, being similar for doped and pure cells, largely cancels in $\Delta\Delta H$.

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